MD Shahriar Hossain

CURRICULUM VITAE

Software Engineer | Full-Stack & AI Systems

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SUMMARY

Computer Science graduate with practical software development experience gained through independent projects and delivering client-facing software solutions. Strong ownership mindset with the ability to take responsibility from understanding requirements through implementation, problem-solving, and delivery. Much of my technical growth has come through self-directed learning beyond the academic curriculum, allowing me to adapt quickly to unfamiliar technologies, tools, and problem domains. Comfortable working independently as well as collaborating with developers and clients. Seeking opportunities where I can contribute across different areas of software engineering while continuing to grow with the needs of the team and product.

PROFESSIONAL EXPERIENCE

Software Engineer — SequenceIT

Jul 2026 – Present

- Develop and deliver full-stack client solutions, contributing to requirement discovery, implementation, and production deployment.
- Built client solutions including **EcomPilot** (AI operations platform), **N Collections** (e-commerce brand), and **LLC Publication Online** (legal compliance engine).
- Guide junior sales lead collection team members on technical feasibility and service capabilities, helping bridge client requirements and technical solutions.

EDUCATION

B.Sc. in Computer Science — BRAC University

2026

CGPA: 3.31 / 4.00

TECHNICAL EXPERTISE

Languages	JavaScript (ES6+), TypeScript, Python(PyTorch, TensorFlow, Scikit-learn), PHP, C, SQL, HTML5, CSS3
Backend & APIs	Node.js, Express.js, Next.js, RESTful APIs, JWT, Redis
Frontend	React, Next.js, TanStack Query, Tailwind CSS, shadcn
Databases &s	PostgreSQL (Supabase RPCs, RLS), MySQL , MongoDB, Mongoose, Prisma
DevOps & Tools	GitHub, Docker, Firebase Auth, Vercel, Render, Postman, jest, Linux

PRODUCTION & CLIENT PROJECTS

EcomPilot — Multi-Marketplace Dashboard & Autonomous AI Agents

GitHub

React 19 · Node.js · Express 5 · MongoDB · OpenAI Responses & Agents API · ReAct Architecture · Traefik · Docker · Vercel

- Architected an asynchronous AI market-discovery engine responding with HTTP 202 while background workers orchestrate a 2-stage search funnel, cutting item evaluation time by estimated ~50% with incremental MongoDB state persistence progress and orphan runs are marked failed on startup.
- Engineered an autonomous Customer Support AI Agent with strict read-only tool gating, thread-level scoping across eBay, TikTok Shop, and Temu, and zero-token deterministic guardrails blocking prompt injections and legal disputes.
- Designed a 3-axis RBAC security model (per-shop sub-admin grants, route permissions, and per-record automatic query scoping) backed by a 30s in-memory sliding-window cache to eliminate redundant database round-trips.
- Built end-to-end commerce operations: multichannel inventory logs, landed-cost profit calculators, CSV/Excel exports with formula-injection sanitization, and analytics powered by Mongo aggregations.

PRODUCTION & CLIENT PROJECTS

N Collections — Boutique E-Commerce Platform

Live Site · GitHub

Next.js 16 (App Router) · React 19 · TypeScript · Supabase (Postgres/Auth/Storage) · Tailwind CSS v4 · Zod · Vercel

- Designed and implemented full-cycle e-commerce platform for a fashion brand: catalog browsing, cart with 45-minute server-side stock reservations, Cash on Delivery/bKash checkout, printable invoices, and public tracking.
- Engineered atomic stock operations via Postgres stored procedures (RPCs) guaranteeing race-safe inventory decrements; secured all endpoints with Row-Level Security (RLS) policies and Zod schema validation.
- Delivered comprehensive administrative backoffice featuring real-time analytics, order lifecycle pipelines, payment verification, product CRUD, and edge-function automated media compression.

LLC Publication Online — Statutory Legal Compliance & Publishing Engine

Live Demo · GitHub

React 19 (Vite) · Node.js · Express · MongoDB Atlas · Mongoose · JWT · Bcrypt · Bilingual i18n · Vercel

- Built statutory tariff resolution platform satisfying NY LLC Law § 206 across all 62 NY counties with foreign/domestic entity calculation rules and server-side pricing verification against DB records.
- Implemented a full bilingual English/Spanish localization architecture with zero-flicker client switching and persistent state.
- Optimized backend routes for serverless cold-starts via cached Mongoose connection pooling and sub-15ms preflight handling.

SurveyHub — AI-Powered Survey & Community Monetization Platform

Live Demo · GitHub

React 19 · Node.js · Express · MongoDB · TanStack Query v5 · BullMQ · Redis · Stripe · Gemini AI · Firebase Auth · Vite

- Architected a resilient 3-provider AI fallback chain (Gemini → OpenRouter → OpenCode Zen) with dynamic rotation for survey analysis and pre-publish automated content moderation.
- Integrated Stripe Checkout and webhook listeners for a credit-wallet monetization system with atomic deductions and role transitions.
- Built a BullMQ and Redis pipeline for automated deadline expiration, paired with factory-pattern RBAC middleware guards and optimistic mutations.

Medix — Pharmacy POS & Inventory Management Platform

GitHub

Next.js 16 (App Router) · React 19 · TypeScript · MongoDB Atlas · Mongoose · Turbopack · useSyncExternalStore

- Developed a pharmacy POS and inventory workflow against a 21k-row catalog with offline localStorage caching, dynamic stock adjustments, and auto-sync to MongoDB Atlas.

BizNest — Multi-Vendor E-Commerce Marketplace

Live Demo · GitHub

React · Node.js · Express · MongoDB · Mongoose · JWT · TanStack Query · Stripe · Helmet.js

- Built a multi-tenant marketplace with strict RBAC supporting Customer, Seller, and Administrator workflows with decoupled Facade custom hooks and TanStack Query caching.
- Hardened backend security with Helmet.js, rate limiting, and server-side Stripe price recalculation against database records to prevent cart tampering.

MedCare Portal — Patient Management & Doctor Appointment Booking

GitHub

Next.js 16 (App Router) · React 19 · TypeScript · Tailwind CSS v4 · Radix UI · shadcn · Sonner

- Built an appointment booking platform with deterministic 15-minute slot grids, queue offset tracking, printable A4 serial tickets, and specialty-filtered doctor discovery.

Grocery Hub — Normalized Relational Database E-Commerce

GitHub

PHP · MySQL · Relational Database Design (3NF) · SQL Indexes & Constraints

- Modeled and normalized a 10-table relational schema in 3NF with 7 foreign key constraints, index tuning, and integrity rules.
- Implemented full 3-role platform (Customer, Seller, Admin) handling product moderation workflows, transaction logging, and a loyalty points engine.

RESEARCH & MACHINE LEARNING

Multi-Omics Sarcoma Subtype Classification & Prognosis

GitHub

Python · PyTorch · Scikit-learn · Gated Attention · Denoising Autoencoder (DAE) · MLP · TCGA Data

- Formulated and trained deep learning architecture combining Denoising Autoencoders with Gated Attention mechanisms for sarcoma subtype classification on genomic/clinical multi-omics data.
- Solved severe class imbalance and low-sample size constraints in TCGA datasets through stratified sampling, specialized normalization pipelines, and prognostic feature discovery.